



1 TITLE AND BRIEF DESCRIPTION OF THE IDEA / MODEL

Project Title: ReportXpert: Autonomous Institutional Intelligence, Accreditation & Reporting Engine

Executive Summary: Universities spend months compiling mountains of documentation across departments every semester for accreditation, ranking audits, and regulatory inspections. **ReportXpert** is an on-premise, sovereign artificial intelligence platform running directly on the university's own **NVIDIA DGX supercomputer node**. It deploys a personalized, fine-tuned Large Language Model (LLM) dedicated exclusively to the university's internal data ecosystem.

The system acts as an ultra-fast digital administrative specialist. It automatically ingests raw institutional files—such as faculty CVs, published papers, patent certificates, grant award letters, student records, and placement data—verifies facts against official citation indices (e.g., Scopus), calculates statutory ratios, audits mathematical accuracy, and produces submission-ready reports for **NAAC SSR, NIRF, MDRF, and internal appraisals in under 5 minutes**. Because all computation occurs locally on campus hardware without internet dependency, confidential student marks, faculty salaries, and unpublished patent drafts remain 100% private with guaranteed zero data leakage.

2 KEY INNOVATIVE FEATURES AND UNIQUENESS

100% Sovereign Privacy Architecture

Zero Cloud Egress

Commercial cloud AI tools (e.g., ChatGPT) transmit prompts and documents to overseas servers, risking legal and regulatory violations. ReportXpert operates strictly within the university's local network on on-premise NVIDIA DGX hardware, ensuring **0.00 KB cloud egress** and absolute institutional data ownership.

Automated Faculty Research Profiler

CV to Proof

Ingests unstructured faculty resumes (Word/PDF) and web profiles. Automatically categorizes research publications, verifies DOIs against Scopus, maps sanctioned government grants (DST, DBT, CSIR), catalogs granted patents, and tallies Academic Performance Indicator (API) appraisal scores.

Dual-AI Engine (Assistant + Auditor)

Cooperative AI

Employs a two-agent architecture: an *Intelligent Coordinator* (Llama 3.2) that communicates in plain everyday English with professors and deans, paired with an independent *Strict Mathematical Auditor* (DeepSeek-R1) that verifies formulas, audits data consistency, and flags anomalies.

Dynamic Multi-Format Template Engine

Word & Excel

Directly outputs official pre-formatted Microsoft Word (`.docx`) books with official letterheads and tables of contents, as well as multi-sheet Microsoft Excel (`.xlsx`) workbooks matching government portal column requirements with zero manual re-formatting required by office staff.

Uniqueness: Why ReportXpert Outperforms Generic Tools

Institutional Sovereign Enclave

On-Premise DGX Compute

Runs locally on high-performance university hardware; requires zero internet connectivity.

Ground-Truth Auditing

Every extracted statistic is tied directly to clickable digital proof files for inspection teams.

Natural Language Querying

Leadership queries: "Compile Criterion 3 research summary" and receives instant verified books.



3 PROBLEM ADDRESSED / PRACTICAL APPLICATION

The Institutional Pain Point: Accreditation and ranking exercises (NAAC, NIRF, MDRF, NBA) are notorious for causing administrative gridlock in higher education institutions:

- **Faculty Drain:** 50+ senior professors and staff are pulled away from core teaching and research for 3 to 4 months every cycle to manually collect certificates, verify publication records, and copy-paste numbers into complex spreadsheets.
- **Costly Discrepancies:** Manual data consolidation results in calculation errors, missed research papers, and misplaced proof documents, directly leading to penalized accreditation grades and lost funding.
- **Privacy Vulnerability:** Uploading institutional data to third-party public web AI engines creates severe risks of confidential exam marks, personnel salaries, and proprietary patent disclosures being leaked externally.

ADMINISTRATIVE DIMENSION	TRADITIONAL MANUAL PROCESS (MONTHS OF STRESS)	WITH REPORTXPERT (AUTONOMOUS & VERIFIED)
Faculty Time & Workload	Months of overtime manually chasing paper receipts & typing tables	Completed in under 5 minutes by the on-premise AI engine
Research & Grant Tracking	Disjointed department files; uncounted citations & missed patents	100% of publications, patents & grants automatically mapped from CVs
Inspection Day Readiness	Panic searching through filing cabinets for misplaced proof files	Instant digital audit trail with 1-click links to every original document
Data Security & Privacy	Severe breach risks by utilizing third-party web tools or external vendors	100% confidential and secured on the university's internal server room

Practical Campus Application: Deans, department heads, and Internal Quality Assurance Cells (IQAC) utilize the conversational console to ask natural-language questions (e.g., "Compile Criterion 3 research dossier for 2024" or "Calculate NIRF student-to-faculty ratio"). The AI aggregates ground-truth records, audits mathematical ratios, and generates ready-to-submit Word dossiers and multi-tab Excel workbooks in seconds.

4 FUTURE POTENTIAL AND SCOPE FOR FURTHER DEVELOPMENT

1. Multi-Campus Federated Intelligence
Architectural capability to link multiple affiliated colleges and regional branch campuses under a unified, federated governance model—enabling central university leadership to aggregate institutional analytics without centralizing raw document storage.

2. Predictive Accreditation Simulation
Integrating proactive predictive models that simulate prospective NIRF ranking shifts and NAAC score changes before final submission. The system will recommend targeted faculty recruitment, citation goals, and research grants needed to attain higher grade tiers.

3. Direct Regulatory API Gateway
Establishing direct, cryptographically authenticated digital submission pipelines to government portals (NAAC portal, AISHE, NIRF API), enabling 1-click end-to-end digital compliance as national accreditation bodies adopt direct API integration.

4. Multilingual & Regional Language Support
Expanding localized speech-to-text and multilingual NLP to support state universities and regional vernacular institutions across India, allowing administrative personnel to query and generate institutional documents in Hindi and regional languages.

Deployment Enclave: Private University Server Room (NVIDIA DGX Compute) **Data Policy:** 100% Local Inference • Zero External Cloud Dependency **Hackathon:** SIH Problem Statement SIH26117